Sparse Tensor Model-Based Spectral Angle Detector for Hyperspectral Target Detection

Jiang Zeng and Qunming Wang

***Abstract*— In recent years, the hyperspectral target detection technique has received widespread attention, and various methods have been proposed. However, most of these methods suffer mainly from two problems. First, the methods perform detection based on the original image, and thus, the targets are easily contaminated by the complex background. Second, a large amount of prior information is usually required, but difficult to obtain. To solve these problems, we proposed a sparse tensor model-based spectral angle (STSA) detector for hyperspectral target detection. First, with a tensor decomposition model, we obtained a sparse tensor component separated from the original image. Based on the sparse tensor, the detection is less contaminated by the background, and the sparse tensor retains the spatial structure using the developed 3-D tensor-based decomposition model. Second, to further enhance the performance, the projected spectral angle method was developed, which only requires a single target spectrum as prior information. Finally, to increase the separation ability between the target and background, we applied a statistical strategy to highlight the target and suppress the background. The experiments were carried out based on four public hyperspectral datasets. The results showed that the proposed STSA method is more accurate than the 11 benchmark methods.**

***Index Terms*— Hyperspectral image (HSI), sparse tensor, spectral angle, target detection.**

I. Introduction

HYPERSPECTRAL images (HSIs) have the advantage of high spectral resolution, usually containing hundreds of spectral bands, presenting an almost continuous spectral curve for each pixel [1]. Therefore, compared to multispectral data, HSIs have received increasing attention in land cover mapping, such as target detection [2], [3], spectral unmixing [4], [5], [6], change detection [7], and image classification [8], [9]. Target detection, as an important application field of HSIs, has urgent requirements in some special fields, such as mineral detection [10], military detection [11], [12], and vegetation detection [13]. Therefore, a number of target detection methods have been developed. According to the principles, target detection methods are roughly divided into the following three categories.

The first category is space-based methods. The spectral angle mapper (SAM) [14] is the basic method of this type, and it determines whether the tested pixel is a target by calculating the projective angle between the tested pixel and the prior spectrum. Based on binary hypothesis testing, the matched filter (MF) and the generalized likelihood ratio test (GLRT) [15] were proposed. MF determines whether the tested pixel is a potential target by calculating the Mahalanobis distance from the tested pixel to the target and background pixels. The formula of the MF is similar to the classic constrained energy minimization (CEM) method [16]. When the MF does not eliminate the first-order statistics of the data, the MF is equivalent to CEM. However, the difference is that CEM is based on the principle of minimizing the total energy of the image to calculate the projective vector. In [17], the orthogonal subspace projection (OSP) suppresses uninteresting pixels by subspace mapping and highlights only the pixels that are close to the prior target of interest. OSP has also been used in other methods as an improvement strategy. For example, the OSP-based GLRT (OSP-GLRT) method [18] projects the detection into the OSP subspace when some background information can be acquired. Meanwhile, it is worthwhile to mention that some methods can also be regarded as spatial projection-based methods from the formula. For example, CEM can be regarded as a distance metric method based on the projecting subspace of eliminating the second-order statistics. The adaptive coherence/cosine estimator (ACE) [19] can be regarded as the spectral angle method based on the semicovariance matrix. Different from the above mapping methods, the kernel function-based methods transform data to a higher dimensional space, where targets are detected. Three kernel functions are commonly used, namely, the radial basis function, polynomial function, and sigmoid function. Accordingly, some methods have been extended to kernel versions, such as kernel matched filter (KMF), kernel constrained energy minimization (KCEM), and kernel orthogonal subspace projection (KOSP).

The second category is representation-based methods, which mainly include model reconstruction and joint representation. Specifically, in [20] and [21], the sparse representation detector (SRD) was first applied to hyperspectral target detection. The premise of SRD is that the backgrounds are sparsely represented by the background dictionary, while the targets are sparsely represented by the target dictionary. It has three versions, including SRD, local SRD, and joint SRD. Amongst them, SRD needs to construct dictionaries of both target and background, and LRD constructs only the target dictionary. Different from the former two versions, JSRD introduces the spatial relationship into SRD. In [22] and [23],

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the sparse representation-based binary hypothesis (SRBBH) and combined sparse and collaborative representation (CSCR) detectors modify the basic assumption in SRD-based methods. SRBBH considers the ubiquity of subpixel targets in mixed pixels, and thus, the premise is modified in such a way that the target can be sparsely represented by the joint dictionaries of the target and background. For CSCR, it is performed based on the assumption that the target can still be sparsely represented by the target dictionary, but the background needs to be linearly represented by the background dictionary, and the strategy of dual windows is used to construct the background dictionary. In [24], the match shrunken cone detector (MSCD) was proposed, which constrains the representation coefficients to the non-negative based on the physical significance of the coefficients. In recent years, many scholars have also focused on strategies based on feature reconstruction. The sparse and low-rank matrix decomposition (SLRMD) method proposed in [25] uses a low-rank and sparse matrix decomposition method to decompose the original image into two parts and completes the detection based on the reconstructed sparse part. Meanwhile, a double-dictionary strategy was proposed in [26] and a tensor model was proposed in [27], [28], and [29], which improve the reconstruction capability of sparse part. Different from feature reconstruction, spatial reconstruction focuses on reconstructing the homogeneity of the neighborhood space. For example, in [30], a total variation-based model is applied to target detection. The decomposition model based on the background dictionary learning (DM-BDL) method proposed in [31] uses a dual strategy of background reconstruction and total variation to increase the accuracy of detection.

The third category is machine learning-based methods, which regard target detection as an information retrieval (IR) problem [32]. Compared to the methods mentioned above, machine learning-based methods weaken the model assumptions to a certain extent. In recent years, machine learning-based methods have been proposed continuously. For example, the asymmetric weighted logistic metric learning (AWLML) proposed in [33] uses the strategy of metric learning to learn a symmetric positive semidefinite matrix for constructing the mapping subspace. In [34], a tree structure-based method determines whether the tested pixel is the target from the binary classification result and the depth of the discriminant. In [35] and [36], multi-instance learning was used to learn the target spectrum by positive and negative packages. To further mine the inherent feature of HSIs, deep learning-based models have been applied to hyperspectral target detection, which shows great potential in data mining and extraction [37]. Common examples include autoencoder (AE) [38], [39], generative adversarial network (GAN) [40], and convolutional neural network (CNN) [41].

In general, the methods mentioned above have different advantages for specific applications, but there are two main problems. First, the accuracy of the methods is easily affected by the complex background of the original HSI, and thus, the detection performance is not robust. Second, some methods require considerable prior information, but it is always difficult to obtain a large amount of prior information, which greatly limits their application ability.

To solve the above problems, we propose the Sparse Tensor Model-Based Spectral Angle (STSA) detector for hyperspectral target detection. Based on the sparse characteristics of the targets (i.e., with small size and low occurrence probability), the STSA obtains the sparse tensor by 3-D tensor decomposition from the original HSI. Compared to the original HSI, the sparse tensor is simpler because it removes a large number of low-rank backgrounds in the HSI. Meanwhile, the targets in the sparse tensor are detected by the mapping-based spectral angle. Last, to cope with the contamination by the background and the noise in the sparse tensor, a statistical strategy is developed to enhance the separating capacity of the target and background. The main contributions are summarized as follows.

1) A 3-D sparse model with a mapping-based spectral angle strategy was proposed, which not only alleviates the interference of low-rank backgrounds, but also increases the detection accuracy in the mapping space.
2) A simple and effective statistical strategy was used, which suppresses the non-targets and separates the backgrounds and targets reliably.
3) Compared with other methods which only need a target spectrum as the additional input data, the proposed STSA method is more robust when the prior spectrum or original image is contaminated by noises.

The remainder of this article is organized as follows. Section II provides a detailed description of the proposed STSA method. Section III presents the experimental results based on four hyperspectral datasets, including the analysis of the proposed STSA method and 11 benchmark methods. In Section IV, we discuss several key issues, followed by the conclusion in Section V.

II. Methods

The flowchart of the proposed STSA method is shown in Fig. 1. The input data include an HSI and a prior target spectrum. The detection process can be divided into the following three parts. First, the HSI data are decomposed into sparse tensors and low-rank tensors through a constructed 3-D tensor model. Second, referring to the prior spectrum, the tested pixels are identified based on the proposed sparse tensor-based spectral angle strategy, and the influence of partial noise and background information is eliminated by the proposed mapping strategy. Finally, a statistical strategy is proposed to highlight the targets and suppress the backgrounds. The proposed STSA method is detailed in Sections II-A.C.

A. Sparse Feature Extraction Based on the Tensor RPCA Model

As a low-rank and sparse decomposition model, robust principal component analysis (RPCA) [42] can separate the original data into the sparse and low-rank parts. However, the limitation is that it can only handle 2-D data. For HSIs with a large number of channels, when using RPCA, it is necessary to convert the 3-D tensor into a 2-D matrix for processing. Such a process usually leads to information loss. In addition, the HSI data contains a large amount of hidden geospatial structural

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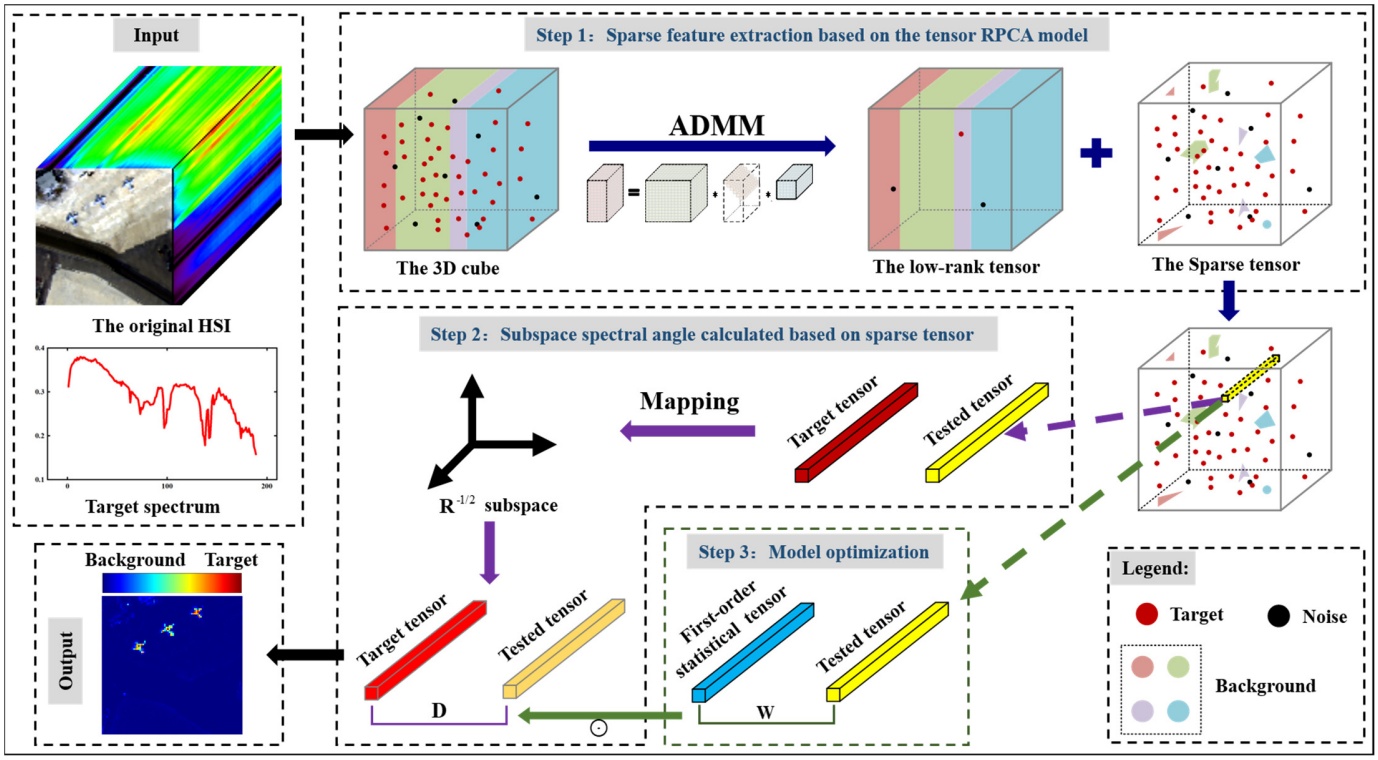


Fig. 1. Flowchart of the proposed STSA-based target detection method.

information, and the neglect of such information would deteriorate the performance in target detection. Alternatively, if a pixel is regarded as a tensor, the spatial relationship between the 3-D data can be regarded as the 2-D relationship of the tensors. As a result, the spatial relationship in the data can be maintained.

In this article, we construct a tensor RPCA model to obtain the sparse tensors. The tensor RPCA model assumes that the original HSI data are a 3-D cube $\mathcal{M} \in \mathbb{R}^{n_1 \times n_2 \times b}$, where n_1 and n_2 are the length and width of the single band image, respectively, and b is the number of bands of the original HSI data. Its mathematical model is as follows:

$$\mathcal{M} = \mathcal{L} + \mathcal{S} \tag{1}$$

where $\mathcal{L}$ and $\mathcal{S}$ are the low-rank tensors and the sparse tensors obtained by decomposition, respectively. Equation (1) is transformed into a directly solvable convex optimization minimum constraint

$$\min_{\mathcal{L},\mathcal{S}} \|\mathcal{L}\|_* + \lambda \|\mathcal{S}\|_1, \quad \text{s.t.} \quad \mathcal{M} = \mathcal{L} + \mathcal{S} \tag{2}$$

in which $\|\cdot\|_*$ is the nuclear norm, $\|\cdot\|_1$ is the l_1-norm, and λ is the penalty coefficient controlling the sparse tensor.

In this article, the augmented Lagrange multiplier method is used to convert (2) into the following augmented Lagrange function:

$$L(\mathcal{L},\mathcal{S},\mathcal{X},\mu) = \arg\min_{\mathcal{L},\mathcal{S}} \|\mathcal{L}\|_* + \lambda\|\mathcal{S}\|_1 + \langle \mathcal{X}, \mathcal{L}+\mathcal{S}-\mathcal{M}\rangle + \frac{\mu}{2}||\mathcal{L}+\mathcal{S}-\mathcal{M}||_F^2 \tag{3}$$

where $\mathcal{X}$ is the Lagrange multiplier tensor, and μ is the tradeoff parameter.

The four variables $\mathcal{L}$, $\mathcal{S}$, $\mathcal{X}$, and μ cannot be solved directly based on $L(\mathcal{L},\mathcal{S},\mathcal{X},u)$. Therefore, the alternating direction method [43] is used to consider each variable in turn. Specifically, (3) is decomposed into four subproblems, and the steps are as follows.

1) Fix Other Variables to Update $\mathcal{L}$: The solution for $\mathcal{L}$ can be transformed into the following subproblem:

$$\mathcal{L}_{k+1} = \arg\min_{\mathcal{L}} \|\mathcal{L}_k\|_* + \frac{\mu_k}{2}\left|\left|\mathcal{L}_k + \mathcal{S}_k - \mathcal{M} + \frac{\mathcal{X}_k}{\mu_k}\right|\right|_F^2 \tag{4}$$

where k is the number of iterations. Equation (4) can be solved by the tensor singular value thresholding method [44].

2) Fix Other Variables to Update $\mathcal{S}$: The solution for $\mathcal{S}$ can be transformed into the following subproblem:

$$\mathcal{S}_{k+1} = \arg\min_{\mathcal{S}} \lambda\|\mathcal{S}_k\|_1 + \frac{\mu_k}{2}\left|\left|\mathcal{L}_{k+1} + \mathcal{S}_k - \mathcal{M} + \frac{\mathcal{X}_k}{\mu_k}\right|\right|_F^2. \tag{5}$$

Equation (5) can be solved by the shrink operator [45].

3) Fix Other Variables to Update $\mathcal{X}$: The solution for $\mathcal{X}$ can be transformed into the following subproblem:

$$\mathcal{X}_{k+1} = \mathcal{X}_k + \mu_k(\mathcal{L}_{k+1} + \mathcal{S}_{k+1} - \mathcal{M}). \tag{6}$$

4) Fix Other Variables to Update μ: The solution for μ can be transformed into the following subproblem:

$$\mu_{k+1} = \min(\sigma\mu_k, \mu_{\max}) \tag{7}$$

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in which σ is the expansion factor, and $\mu_{\max}$ is the maximum tradeoff parameter of μ. When the iterative reaches the maximum value $T_{\max}$ or satisfies the convergence condition defined by the following equation, the iterations end:

$$\max\left(||\mathcal{L}_{k+1}-\mathcal{L}_k||_\infty, ||\mathcal{S}_{k+1}-\mathcal{S}_k||_\infty, ||\mathcal{L}_{k+1}+\mathcal{S}_{k+1}-\mathcal{M}||_\infty\right) < \varepsilon \quad (8)$$

where $\max(\cdot)$ is the notation for the maximum value, $\|\cdot\|_\infty$ is the Frobenius norm, and ε is the discriminant factor.

According to the above four subproblems, the original image $\mathcal{M}$ can be decomposed into a low-rank tensor part $\mathcal{L}$ and a sparse tensor part $\mathcal{S}$.

B. Subspace Spectral Angle Calculated Based on the Sparse Tensor

In the sparse tensors $\mathcal{S}$ obtained by the tensor RPCA model, although a large number of complex low-rank backgrounds are removed, a small amount of noise and background information is still retained. Therefore, the next processing has two goals: target matching and interference suppression.

The spectral angle is a classic metric for measuring the spectral difference. In hyperspectral target detection, it determines whether the tested pixel is a target pixel by calculating the cosine of the spectral angle between the tested pixel and the target spectrum. However, the disadvantage of the spectral angle is the anti-interference ability. In this article, the spectral angle is calculated based on the sparse tensor part of the original image, which largely eliminates the influence of complex backgrounds. To further enhance the detection performance, a mapping-based spectral angle is applied [18]. The specific details are as follows:

Assuming that the sparse tensor $\mathcal{S}$ is a 3-D cube with a size of $n_1 \times n_2 \times b$, and it is converted into a 2-D matrix $\mathbf{S}$ with a size of $n_1 n_2 \times b$. The formula of mapping matrix $\mathbf{R_S}$ of $\mathbf{S}$ is as follows:

$$\mathbf{R_S} = \frac{1}{n_1 n_2}\left(\mathbf{S}^{\mathrm{T}}\mathbf{S}\right) \quad (9)$$

in which $\mathbf{R_S}$ is the second-order statistical covariance matrix with a size of $b \times b$. The diagonal values of $\mathbf{R_S}$ are the second-order origin moment of each band. The mapping matrix used to eliminate the second-order statistics of the sparse matrix is as follows:

$$\mathbf{R_S}^{-1/2} = \mathbf{Q}\mathbf{V}^{-1/2}\mathbf{Q} \quad (10)$$

in which $\mathbf{R_S}^{-1/2}$ is the affine matrix used for the map, $\mathbf{V}$ is the eigenvalue diagonal matrix of $\mathbf{R_S}$, and $\mathbf{Q}$ is the orthogonal identity matrix and its column vector is the eigenvector that corresponds to $\mathbf{R_S}$. By projecting the target spectrum $\mathbf{d}$ and the sparse tensors $\mathcal{S}$ into the $\mathbf{R_S}^{-1/2}$ subspace, the covariance of $\mathcal{S}$ between the bands can be regularized. The formula is as follows:

$$\cos\theta^{\wedge}_{(i,j)} = \frac{\left\langle \mathbf{R_S}^{-1/2}\mathcal{S}_{(i,j,:)}, \mathbf{R_S}^{-1/2}\mathbf{d}\right\rangle}{\left\|\mathbf{R_S}^{-1/2}\mathcal{S}_{(i,j,:)}\right\| \cdot \left\|\mathbf{R_S}^{-1/2}\mathbf{d}\right\|} \quad (11)$$

where (i, j) represents the spatial location of the tested pixel, $\mathcal{S}_{(i,j,:)}$ is the tensor of the tested pixel, and $\cos\theta^{\wedge}_{(i,j)}$ is the mapping-based spectral angle cosine between the tested tensor and the target spectrum. Although the computational complexity of both $\mathbf{R_S}^{-1}$ and $\mathbf{R_S}^{-1/2}$ is $o(b^3)$, $\mathbf{R_S}^{-1}$ is easier to calculate than $\mathbf{R_S}^{-1/2}$. Therefore, to simplify calculation, (11) is converted into the following equation:

$$\mathbf{D}_{(i,j)} = \left[\cos\theta^{\wedge}_{(i,j)}\right]^2 = \frac{\left(\mathcal{S}^{\mathrm{T}}_{(i,j,:)}\mathbf{R_S}^{-1}\mathbf{d}\right)^2}{\left(\mathcal{S}^{\mathrm{T}}_{(i,j,:)}\mathbf{R_S}^{-1}\mathcal{S}_{(i,j,:)}\right)\left(\mathbf{d}^{\mathrm{T}}\mathbf{R_S}^{-1}\mathbf{d}\right)} \quad (12)$$

in which $\mathbf{D}$ is the target detection map obtained by the sparse tensor-based mapping spectral angle method.

C. Model Optimization Based on Statistical Suppression

$\mathbf{D}$ is obtained based on the $\mathbf{R_S}^{-1/2}$ subspace detection under the sparse tensor $\mathcal{S}$. Although $\mathbf{D}$ leads to great detection accuracy, its separation capability of the target and background is still limited. Meanwhile, the background and noise are ubiquitous with low energy relative to targets in the sparse tensor $\mathcal{S}$. That is, the statistical value of the background and noise is closer to the first-order origin moment of the sparse tensor $\mathcal{S}$. Therefore, we propose a statistical strategy by eliminating the first-order statistics of $\mathcal{S}$ to enhance the separation capability between the target and background in $\mathbf{D}$. The detailed formula is as follows.

$$\mathbf{W}_{(i,j)} = \left|\left|\mathcal{S}_{(i,j,:)} - \mathbf{u}_{\mathcal{S}}\right|\right|_2^2 \quad (13)$$

where $\mathbf{u}_{\mathcal{S}}$ is the tensor that shows the first-order origin moment of each band of $\mathcal{S}$, and $\mathbf{W}$ is the statistical weight matrix.

$$\mathbf{D}_{\mathrm{TD}} = \mathbf{W} \odot \mathbf{D} \quad (14)$$

in which $\mathbf{D}_{\mathrm{TD}}$ is the final detection map, and $\odot$ is the Hadamard product operator symbol.

The statistical strategy used to optimize $\mathbf{D}$ in (14) not only keeps the target detection ability of (12) but also enhances the separation ability between the target and background. Meanwhile, the sensitivity for the background and noise based on the single prior spectrum method is weakened. In addition, the consideration of $\mathbf{W}$ enhances the spatial constraint of $\mathbf{D}$.

The entire flow of the proposed STSA method is summarized as Algorithm 1.

III. Experiments

To demonstrate the effectiveness of the proposed STSA method, four public hyperspectral datasets were used in the experiments. The corresponding prior information extracted manually (10 targets and 20 backgrounds) was also used. In addition, a GF-5 dataset covering a large scene was also used for validation. Moreover, the proposed STSA method was compared with the 11 benchmark methods, including MF, GLRT [15], OSP [17], OSP-GLRT [18], CEM [16], tensor principal component analysis (TPCA)-CEM [28], ACE [19], SRBBH [22], CSCR [23], SLRMD [25], and double dictionary-based non-linear representation model for hyperspectral sub-pixel target detection (DDNRTD) [26]. The experimental platform was a computer with a 3.6-GHz CPU, 32-G RAM, and 64-bit Windows 10. All methods were implemented using MATLAB R2020a. Section III-A describes the four public hyperspectral datasets used in the experiments.

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Algorithm 1 The STSA Method

Input:

a) the original HSI data $\mathcal{M} \in \mathbb{R}^{n_1 \times n_2 \times b}$

b) *a priori* target spectral $\mathbf{d} \in \mathbb{R}^{1 \times b}$

Output: The detection map $\mathbf{D}_{TD}$.

1. Initialization:

Set $\mathcal{L}_0 = \mathbf{0}$, $\mathcal{S}_0 = \mathbf{0}$, $\mathcal{X}_0 = \mathbf{0}$, $\mu_0 = 1e-4$, $\sigma = 1.1$, $\lambda = 1/\sqrt{\max(n_1, n_2)b}$ and $\mu_{\max} = 1e10$. Set a tolerance $\varepsilon = 1e-8$ and a maximum number of iterations $T_{\max} = 500$.

2. Sparse feature extraction:

Update the low-rank sensor $\mathcal{L}_{k+1}$ by Eq. (4).

Update the sparse sensor $\mathcal{S}_{k+1}$ by Eq. (5).

Update the Lagrange multiplier sensor $\mathcal{X}_{k+1}$ by Eq. (6).

Update μ_{k+1} by Eq. (7).

If $k < T_{\max}$ or Eq. (8) is satisfied, the sparse tensor $\mathcal{S}$ is extracted.

3. Subspace spectral angle calculation:

Based on the target spectrum $\mathbf{d}$ and the sparse tensor $\mathcal{S}$, produce the detection map $\mathbf{D}$ by Eq. (12).

4. Model optimization:

Compute the statistical weight matrix $\mathbf{W}$ based on the sparse tensor $\mathcal{S}$ by Eq. (13).

Produce the final detection by Eq. (14).

Section III-B presents the evaluation indices. Section III-C analyzes the performance of various methods based on the experimental results. The ablation and robustness analysis are performed in Sections III-D and III-E, respectively. Meanwhile, the application of the STSA method to a larger scene is described in Section III-F.

A. Hyperspectral Datasets

The key information of the four public datasets is shown in Table I, and the details are described below.

1) HSI-I Dataset: The first dataset was acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor. The coverage area is located in an urban area in Michigan, USA [46], which contains 210 bands. The spatial resolution is 2.0-m per pixel, and the spatial size is 80 × 100 pixels. A total of 162 bands were retained after excluding the low signal-to-noise ratio (SNR) and water absorption bands. Nineteen pixels that contain vehicles were selected as targets with sizes between 2 and 4 pixels, and the spatial distribution of the targets is scattered. The true-color images and references are provided in Fig. 2(a) and (b). The prior spectral curves of the targets and backgrounds are provided in Fig. 2(c).

2) HSI-II Dataset: The second dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [47]. The image covers a military area in the city of San Diego, USA, with a spatial resolution of 3.5 m per pixel, and it contains 224 bands. The spatial size is 90 × 90 pixels. A total of 189 bands were retained after excluding the SNR and water absorption bands. The three planes with a total of 77 pixels were selected as the targets. The true-color images and reference of the targets are provided in Fig. 3(a) and (b). The prior spectral curves are provided in Fig. 3(c).

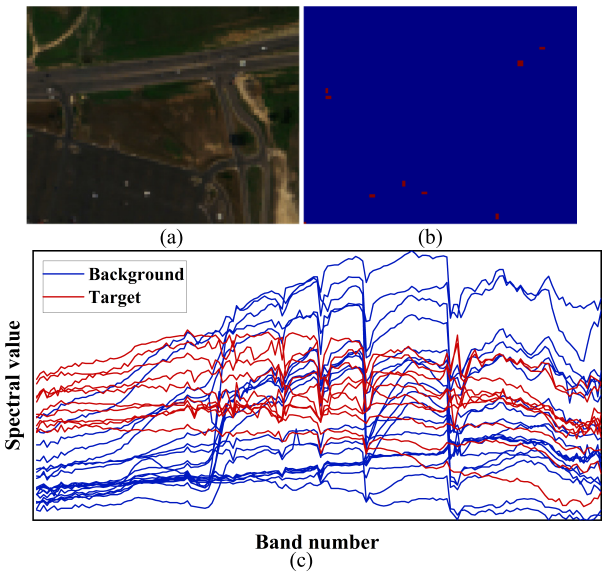


Fig. 2. HSI-I dataset. (a) True-color image of HSI-I. (b) Reference map of the target. (c) Prior spectral curves of the targets and backgrounds.

TABLE I

DETAILS OF THE FOUR DATASETS USED IN THE EXPERIMENTS

Dataset	Sensor	Location	Spatial resolution (m)	Spatial size (pixels)	Target proportion (%)
HSI-I	HYDICE	Michigan	2.0	80′ 100	0.24
HSI-II	AVIRIS	San Diego	3.5	90′ 90	0.95
HSI-III	AVIRIS	San Diego	3.5	100′ 100	1.34
HSI-IV	AVIRIS	Mississippi	3.4	100′ 100	0.6

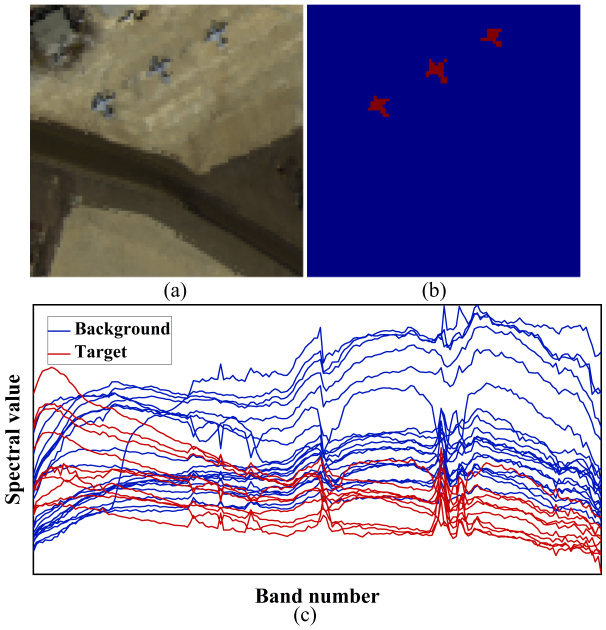


Fig. 3. HSI-II dataset. (a) True-color image of HSI-II. (b) Reference map of the target. (c) Prior spectral curves of the targets and backgrounds.

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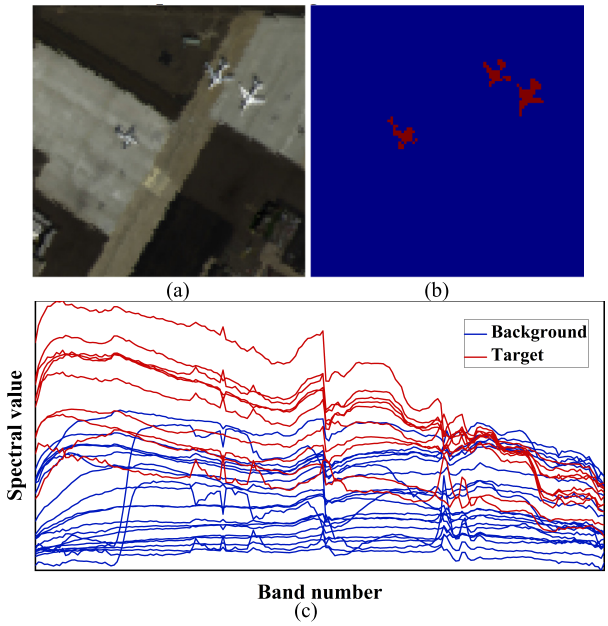


Fig. 4. HSI-III dataset. (a) True-color image of HSI-III. (b) Reference map of the target. (c) Prior spectral curves of the targets and backgrounds.

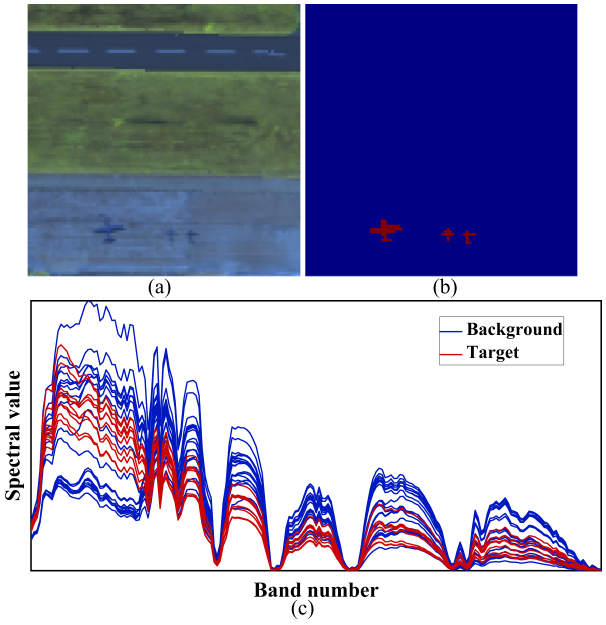


Fig. 5. HSI-IV dataset. (a) True-color image of HSI-IV. (b) Reference map of the target. (c) The prior spectral curves of the targets and backgrounds.

3) HSI-III Dataset: The third dataset is another area acquired under the same conditions as the HSI-II dataset, and the image spatial size is 100 × 100 pixels. The area covers three planes, and 134 pixels were selected as the targets. The corresponding true-color images and references are shown in Fig. 4(a) and (b). Meanwhile, the prior spectral curves are provided in Fig. 4.

4) HSI-IV Dataset: The fourth dataset was acquired with the AVIRIS sensor. The coverage area is located in Gulfport, Southern, Mississippi, USA [48]. The spatial resolution is 3.4 m per pixel, and the wavelength range is between 400 and 2500 nm. After removing the water absorption and low SNR bands, 191 bands were retained. The spatial size of the image is 100 × 100 pixels, and 60 pixels that contain three planes were selected as the targets. The true-color images and reference of the targets are provided in Fig. 5(a) and (b). The prior spectral curves of the targets and backgrounds are shown in Fig. 5(c).

B. Evaluation Indices

The accuracy evaluation indices used are introduced in this section, including the receiver operating characteristic (ROC) curve, the area under the curve (AUC) [49], the normalized AUC (NAUC) [34], the square error ratio (SER) [50], and the box plots of target-background [51].

1) ROC Curve, AUC, and NAUC: The ROC curve is used to represent the relationship between the probability of detection (pd), the probability of false detection (pf), and the discrimination threshold (τ). pd is the probability that the target is detected under a fixed τ. pf corresponds to the probability that a non-target is detected as a target under a fixed τ. The formulas are given in the following equations:

$$\mathrm{pd} = \frac{N_d}{N_{T_{\mathrm{tol}}}} \tag{15}$$

$$\mathrm{pf} = \frac{N_f}{N_{B_{\mathrm{tol}}}} \tag{16}$$

where N_d is the number of target pixels detected as the target, N_f is the number of background pixels detected as target under the fixed τ, $N_{T_{\mathrm{tol}}}$ is the total number of target pixels, and $N_{B_{\mathrm{tol}}}$ is the total number of background pixels. Different thresholds τ lead to different binary results of detection, presenting different pd and pf. In this article, the ROC curve of (pd,pf) was used to evaluate the detection accuracy of the method, and the ROC curve of (pf, τ) was used to evaluate the background suppression accuracy. Quantitatively, $\mathrm{AUC}_{(\mathrm{pd,pf})}$ and $\mathrm{AUC}_{(\mathrm{pf},\tau)}$ are the areas under the ROC curves of (pd, pf) and (pf, τ), which are used for calculating detection and background suppression accuracies. The $\mathrm{AUC}_{(\mathrm{pd,pf})}$ and $\mathrm{AUC}_{(\mathrm{pf},\tau)}$ are calculated as follows:

$$\begin{aligned} \mathrm{AUC}_{(\mathrm{pd,pf})} &= \frac{1}{2}\sum_{i}^{n}\left(\mathrm{pd}_i + \mathrm{pd}_{i-1}\right)\left(\mathrm{pf}_i - \mathrm{pf}_{i-1}\right) \\ \mathrm{AUC}_{(\mathrm{pf},\tau)} &= \frac{1}{2}\sum_{i}^{n}\left(\mathrm{pf}_i + \mathrm{pf}_{i-1}\right)\left(\tau_i - \tau_{i-1}\right) \end{aligned} \tag{17}$$

where $\{\mathrm{pd}, \mathrm{pf}, \tau, \mathrm{AUC}_{(\mathrm{pd,pf})}, \mathrm{AUC}_{(\mathrm{pf},\tau)}\} \in (0, 1)$, τ_i is the ith component when (0, 1) is divided into n parts, pd_i and pf_i are the corresponding calculated values under τ_i.

The overall detection accuracy is calculated through $\mathrm{AUC}_{(\mathrm{pd,pf})}$ and $\mathrm{AUC}_{(\mathrm{pf},\tau)}$ as follows:

$$\mathrm{AUC}_{\mathrm{TD\text{-}BS}} = \mathrm{AUC}_{(\mathrm{pd,pf})} - \mathrm{AUC}_{(\mathrm{pf},\tau)} \tag{18}$$

where $\mathrm{AUC}_{\mathrm{TD\text{-}BS}} \in (-1, 1)$ represents the overall detection accuracy and the larger the value, the greater the overall accuracy.

NAUC is used as a supplementary metric to evaluate the detection performance when the number of targets is far smaller than that of backgrounds. It calculates AUC at a

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TABLE II
OPTIMAL PARAMETER SETTINGS OF THE METHODS

Methods	Parameter	Datasets			
		HSI-I	HSI-II	HSI-III	HSI-IV
SRBBH	w_{in}	9	11	11	11
	w_{out}	11	15	15	15
	k	9	9	9	9
CSCR	w_{in}	7	11	11	11
	w_{out}	9	15	15	15
	k	3	3	3	3
SLRMD	t	1	1	1	1
	l	1	$6e-1$	$6e-1$	$2e-1$
DDNRTD	l	$1e-6$	$2e-3$	$5e-4$	$1e-4$

lower pf (e.g., pf $= 1 \times 10^{-2}$), making the number of targets to be detected close to the number of backgrounds that are incorrectly detected as targets.

2) SER: SER evaluates the detection accuracy by calculating the squared error ratio between the detection map and the reference. The specific formula is as follows:

$$\mathrm{SER} = \frac{\sum_{i=1}^{N_{T_{\mathrm{tol}}}} (p_i - 1)^2 + \sum_{j=1}^{N_{B_{\mathrm{tol}}}} \left(p_j - 0\right)^2}{N_{T_{\mathrm{tol}}} + N_{B_{\mathrm{tol}}}} \times 100 \qquad (19)$$

where p_i is the normalized value of the ith target pixel in the detection map, and p_j is the normalized value of the jth background pixel in the detection map. Ideally, the normalized values of the target and background pixels are close to 1 and 0, respectively. Therefore, the smaller the SER value is, the greater the detection accuracy, and vice versa. Different from the ROC and AUC values that are calculated by selecting different thresholds, SER directly evaluates the overall accuracy according to the residuals between the detection map and the reference.

3) Box Plot Map: The detection map is divided into target and background sets according to the reference, and box plots are generated based on the two types of sets. It can intuitively illustrate the separation ability of the target and background.

C. Detection Performance

To ensure the fairness of the comparison, all methods used the same prior spectra extracted manually, as shown in Figs. 2(c)–5(c). Meanwhile, an average method was used to generate a prior target spectrum when only one prior spectrum is required in the methods (i.e., MF, GLRT, CEM, TPCA-CEM, ACE, and STSA). For the methods (SRBBH, CSCR, SLRMD, and DDNRTD) involving parameters, the optimal parameter settings are given in Table II. For the four datasets, the detection results of the 12 methods under the optimal parameters are shown in Fig. 6.

Fig. 6(a) and (b) shows the true-color images and references of the four datasets, respectively. Fig. 6(c)–(n) shows the detection maps of 12 methods. In the HSI-I dataset, the target pixels have very small sizes, including some subpixel pixels. Generally, four points can be observed in Fig. 6(a1)–(n1). First, the proposed STSA method has the most balanced performance of target detection and background suppression overall compared to the other 11 benchmark methods. Second, OSP, CSCR, SLRMD, and DDNRTD have the second greatest detection performance, but the background suppression ability is relatively limited, leading to a higher pf than other methods. For example, the separation ability of the OSP method is relatively limited, due to the obvious background. In the detection map of SLRMD, a large area of bare ground is highlighted in the lower right of the map, and thus, SLRMD could have false detections when the threshold is high. Third, MF, GLRT, CEM, TPCA-CEM, and ACE show weaker performance than the above four methods. Specifically, some targets are missed, resulting in a low pd, but their background suppression ability is greater than that of OSP, CSCR, and SLRMD. Finally, OSP-GLRT and SRBBH show weaker detection performance among all methods. Specifically, the background is over-suppressed in OSP-GLRT while the background is less suppressed in SRBBH.

For the HSI-II, HSI-III, and HSI-IV datasets, the targets are aircraft, but the difference is the types of aircraft and backgrounds. From Fig. 6(a2)–(n4), it can be seen that the proposed STSA method has the greatest overall detection performance and the background is well suppressed. Second, the targets of the OSP and SLRMD methods are highlighted satisfactorily, but the background suppression is limited compared to other methods. Third, the background suppression of the MF, GLRT, CEM, TPCA-CEM, CSCR, and DDNRTD methods is more satisfactory than OSP and SLRMD but weaker than the proposed STSA method. It can be seen that there are some local background textures, such as the houses in the upper left corner of the HSI-II dataset, and the zebra stripes on the impervious surface in the HSI-IV dataset. Compared with the above methods, the detection map of CSCR is relatively scattered, and the spatial integrity of the map is limited.

Fig. 7 shows the $\mathrm{ROC}_{(\mathrm{pd,pf})}$ curve of (pd, pf). For the HSI-II and HSI-III datasets, the proposed STSA method has a larger pd than the 11 benchmark methods under various thresholds. For the HSI-I dataset, SRBBH has the largest pd when the pf is less than 0.02, but when the pf exceeds 0.04, the proposed STSA method has the largest pd, which is kept at 1 when the false alarm rate is larger than 0.04. For the HSI-IV dataset, when the pf is larger than 0.06, the proposed STSA always maintains the largest pd, and the pd is close to 1 when the pf is larger than 0.1.

Fig. 8 shows the $\mathrm{ROC}_{(\mathrm{pf},\tau)}$ curve of (pf, τ), which is used to reflect the background suppression of the methods. Overall, the proposed STSA method generally has the most satisfactory background suppression ability of the four datasets. In the HSI-III and HSI-IV datasets, the background suppression capabilities of the OSP-GLRT and ACE methods are close to the proposed SAST method, and the pf is close to 0 after the threshold is larger than 0.1. For the HSI-I and HSI-II datasets, OSP-GLRT and ACE have the lowest probability of false when the threshold is less than 0.1. Fig. 9 shows the NAUC values of different methods when pf is less than 1×10^{-2} in the four datasets. It can be seen that the proposed STSA method has the largest NAUC value in HSI-II and HSI-III datasets.

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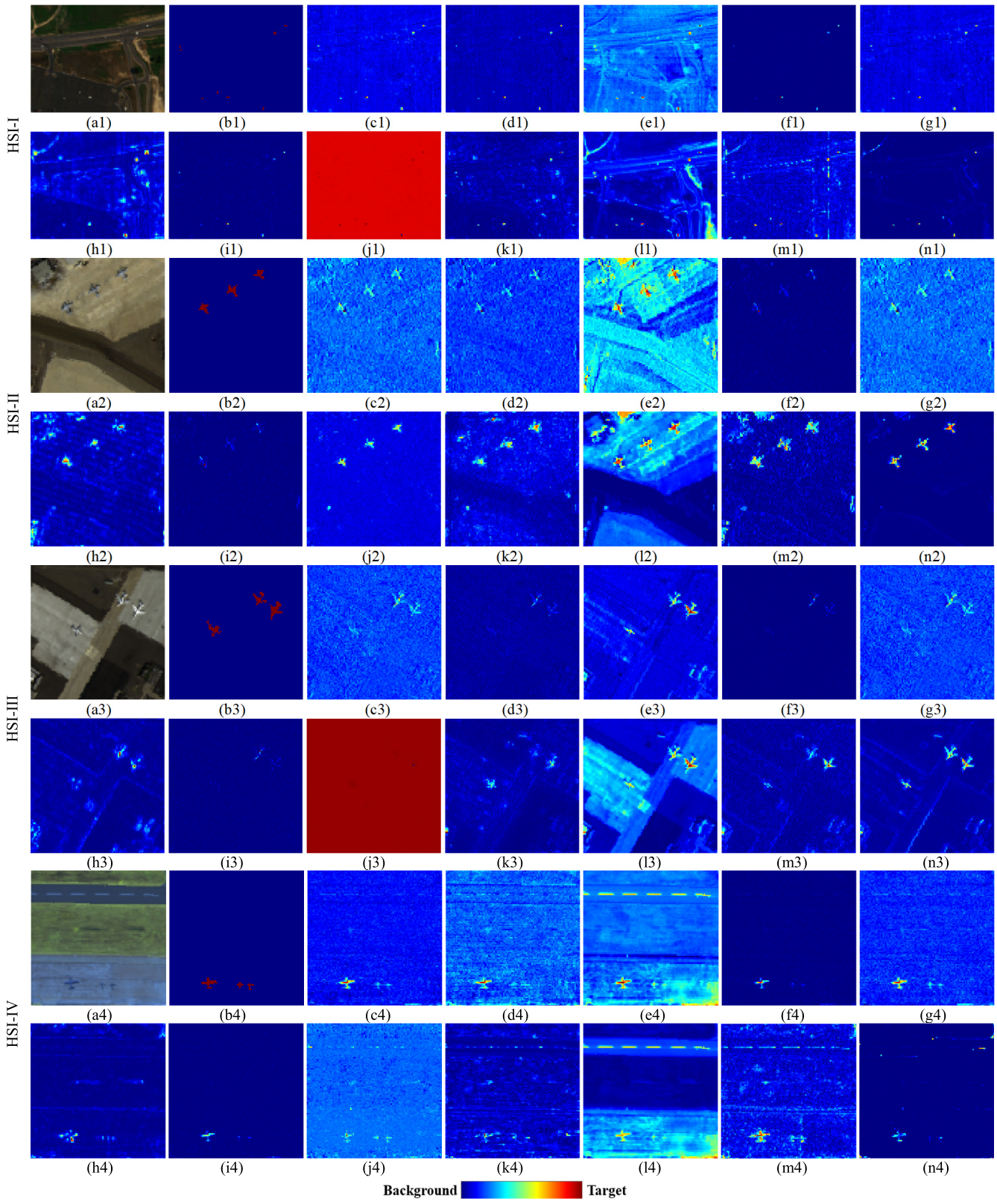


Fig. 6. Detection maps of different methods for the four datasets. (a) Original HSI. (b) Reference of the target. (c) MF. (d) GLRT. (e) OSP. (f) OSP_GLRT. (g) CEM. (h) TPCA-CEM. (i) ACE. (j) SRBBH. (k) CSCR. (l) SLRMD. (m) DDNRTD. (n) STSA.

Fig. 10 shows the box plots of target-background separation for different methods in the four datasets. As shown in Fig. 9, first, the background box ranges of SLRMD are larger than those of the other methods. That is, the background

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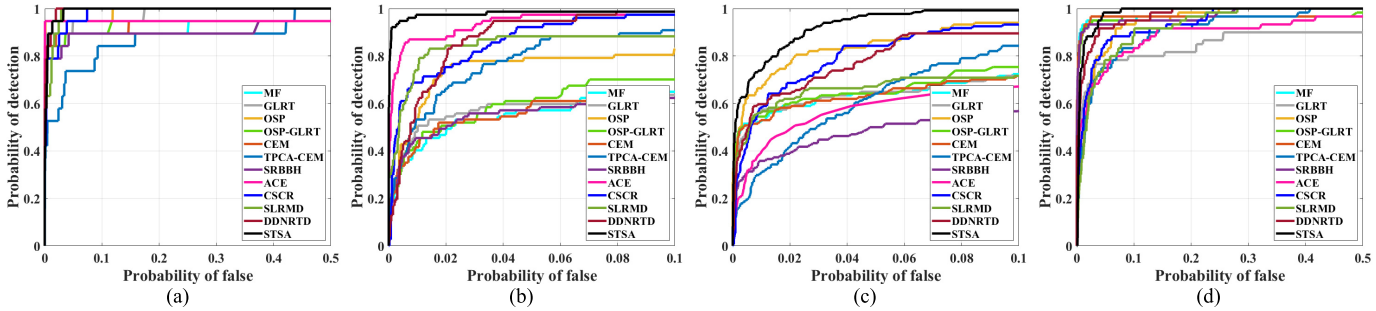


Fig. 7. $\mathrm{ROC}_{(\mathrm{pd,pf})}$ of the detection maps. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

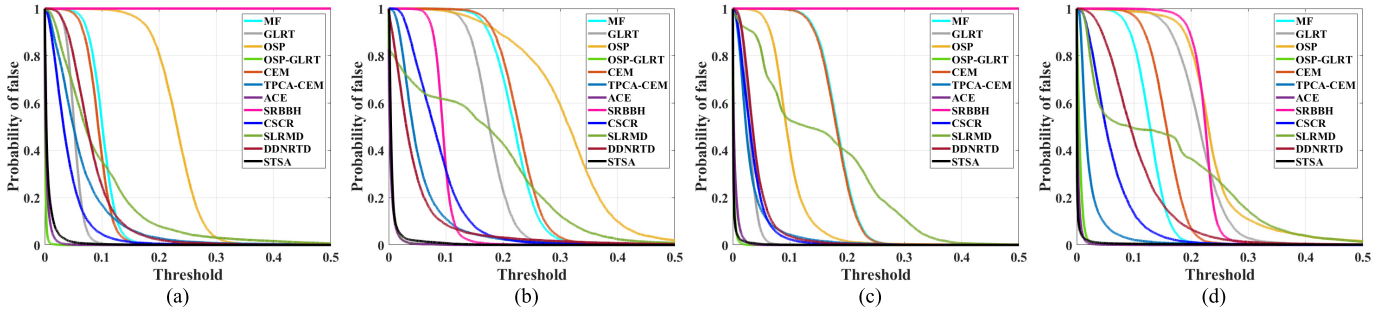


Fig. 8. $\mathrm{ROC}_{(\mathrm{pf},\tau)}$ of the detection maps. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

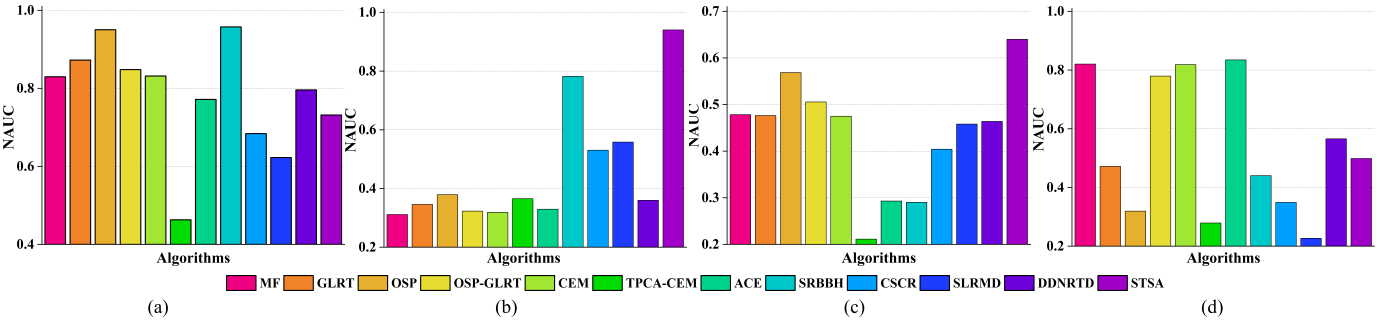


Fig. 9. NAUC of the detection maps. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

suppression is relatively limited. Second, in the HSI-I, HSI-II, and HSI-III datasets, the target boxes of the MF, GLRT, and CEM methods overlap with the background boxes to a certain extent, which indicates that the likelihood of a false result is inevitable. Third, the background-target box spans of SRBBH in the HSI-I and HSI-III datasets are the smallest and closest to 1. In general, in the four datasets, the background box span of the proposed STSA method is the smallest and closest to 0. There is a relatively large span between the target box and the background box. That is, the overall separation ability of the proposed STSA method is greater than that of the other 11 benchmark methods.

The computing time of each method is shown in Table III. It can be seen that STSA has a larger computational cost than that of the other 11 benchmark methods. Table IV shows the accuracy of the 12 methods for the four datasets, including the $\mathrm{AUC}_{(\mathrm{pd,pf})}$, $\mathrm{AUC}_{(\mathrm{pf},\tau)}$, $\mathrm{AUC}_{\mathrm{TD\text{-}BS}}$, and SER values. As shown in Table IV, all evaluation indices show that the proposed STSA method not only ensures the ability of background suppression, but also presents greater detection accuracy than the other 11 benchmark methods.

TABLE III
COMPUTING TIME (IN UNITS OF SECONDS) OF EACH METHOD FOR THE FOUR DATASETS

Methods	Datasets			
	HSI-I	HSI-II	HSI-III	HSI-IV
MF	0.01	0.02	0.02	0.02
GLRT	0.09	0.10	0.16	0.14
OSP	0.00	0.00	0.00	0.01
OSP-GLRT	0.10	0.11	0.16	0.16
CEM	0.01	0.01	0.01	0.01
TPCA-CEM	7.62	9.14	11.53	12.36
ACE	0.87	0.44	0.70	0.63
SRBBH	1.56	8.91	11.71	10.80
CSCR	4.49	13.99	17.78	17.73
SLRMD	1.92	2.60	2.77	2.74
DDNRTD	12.35	29.62	28.24	30.71
STSA	24.54	28.91	35.53	38.58

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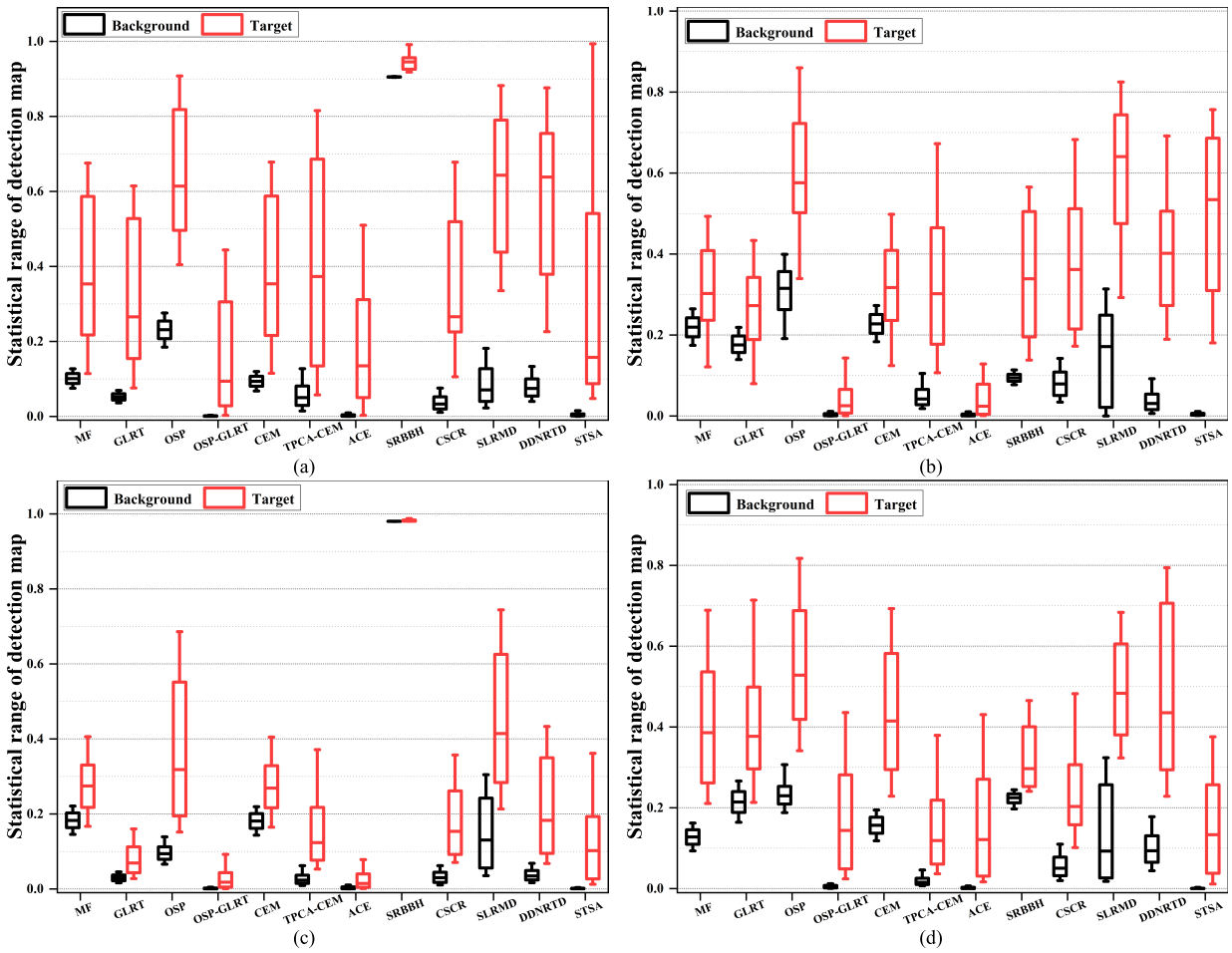


Fig. 10. Box plots of background and target in the detection maps. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

TABLE IV
ACCURACY OF DIFFERENT METHODS FOR THE FOUR DATASETS

Datasets		Methods											
		MF	GLRT	OSP	OSP-GLTR	CEM	TPCA-CEM	ACE	SRBBH	CSCR	SLRMD	DDNRTD	STSA
HSI-I	$AUC_{(pd,pf)}$	0.9399	0.9879	0.9935	0.9442	0.9477	0.9296	0.9378	0.9473	0.9903	0.9936	**0.9967**	0.9958
	$AUC_{(pf,\tau)}$	0.1014	0.0524	0.2303	**0.0013**	0.0942	0.0642	0.0035	0.9053	0.0404	0.0949	0.0825	0.0066
	AUC_{TD-BS}	0.8385	0.9355	0.7632	0.9429	0.8535	0.8654	0.9343	0.0420	0.9499	0.8987	0.9142	**0.9892**
	SER	1.1694	0.4113	5.4811	0.1683	1.0298	0.8538	0.1618	81.7870	0.3800	1.6329	0.9479	**0.1587**
HSI-II	$AUC_{(pd,pf)}$	0.7593	0.7430	0.9042	0.8421	0.7527	0.9635	0.8304	0.9819	0.9822	0.9670	0.9829	**0.9975**
	$AUC_{(pf,\tau)}$	0.2203	0.1781	0.3091	0.0047	0.2287	0.0564	**0.0042**	0.0957	0.0863	0.1574	0.0475	0.0069
	AUC_{TD-BS}	0.5390	0.5649	0.5951	0.8374	0.5240	0.9071	0.8262	0.8862	0.8959	0.8096	0.9354	**0.9906**
	SER	5.4212	3.7879	10.4180	0.8566	5.7836	1.0488	0.8630	1.3721	1.4044	4.2921	1.002	**0.3213**
HSI-III	$AUC_{(pd,pf)}$	0.8471	0.8592	0.9754	0.8698	0.8433	0.9484	0.8083	0.8383	0.9722	0.9111	0.9586	**0.9915**
	$AUC_{(pf,\tau)}$	0.1829	0.0301	0.0996	**0.0018**	0.1810	0.0326	0.0040	0.9803	0.0349	0.1538	0.0410	0.0023
	AUC_{TD-BS}	0.6642	0.8291	0.8758	0.8680	0.6623	0.9158	0.8043	-0.1420	0.9373	0.7573	0.9176	**0.9892**
	SER	4.0937	1.2246	1.6772	1.2426	4.0327	1.1671	1.2554	94.8262	1.1195	3.9551	1.0760	**1.0121**
HSI-IV	$AUC_{(pd,pf)}$	0.9662	0.8922	0.9711	0.9794	0.9738	0.9505	0.9848	0.9310	0.9667	0.9600	0.9839	**0.9907**
	$AUC_{(pf,\tau)}$	0.1282	0.2155	0.2418	0.0058	0.1571	0.0239	**0.0026**	0.2231	0.0605	0.1481	0.1055	0.0033
	AUC_{TD-BS}	0.8380	0.6767	0.7293	0.9736	0.8167	0.9266	0.9822	0.7079	0.9062	0.8119	0.8784	**0.9874**
	SER	1.9460	5.0357	6.4762	**0.4150**	2.7547	0.5748	0.4309	5.2766	0.8897	4.2079	1.6719	0.5135

D. Ablation Analysis

To analyze the effectiveness of the three strategies in the proposed STSA method, a three-part ablation experiment was performed on the HSI-I dataset. Specifically, the detection only performed through the sparse feature extraction in Section II-A is denoted in Part 1. Compared to Part 1, Part 2

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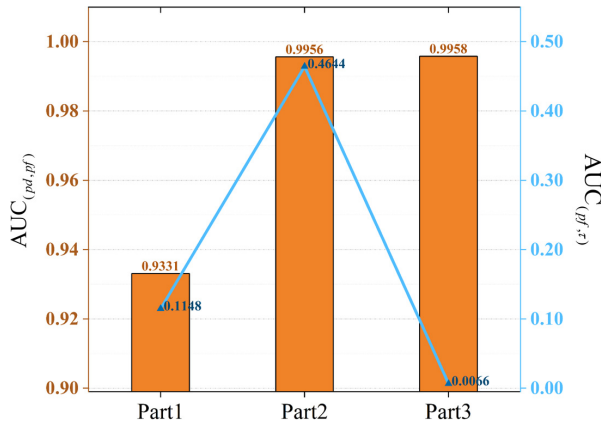


Fig. 11. Bar-lines of $AUC_{(pd,pf)}$ and $AUC_{(pf,\tau)}$ values of each part of the STSA for HSI-I dataset.

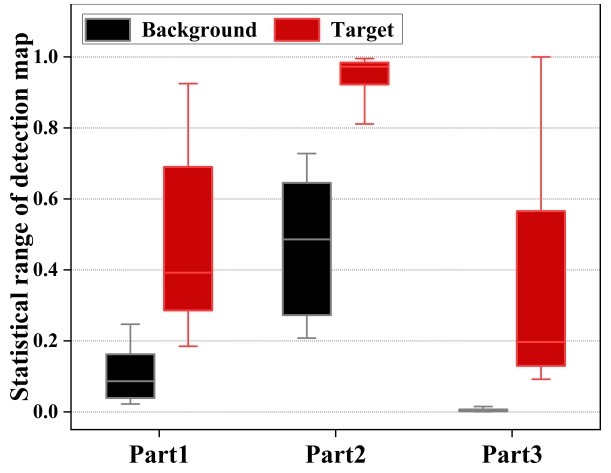


Fig. 12. Box plots of background and target of each part of the proposed STSA method for the HSI-I dataset.

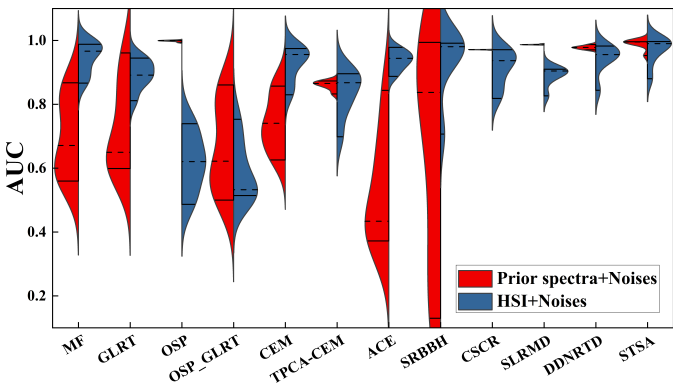


Fig. 13. Violin plots of $AUC_{(pd,pf)}$ for each method when the original image or prior target spectrum was contaminated by different levels of zero-mean Gaussian noises.

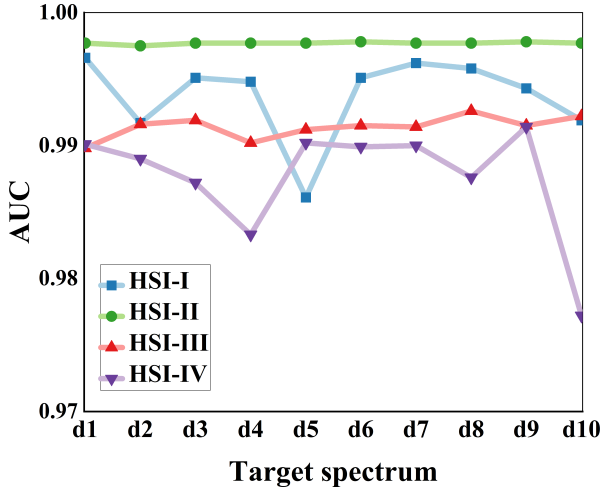


Fig. 14. $AUC_{(pd,pf)}$ of STSA under different prior target spectra for the four datasets.

adds the subspace spectral angle measurement, as illustrated in Section II-B. The complete STSA method is denoted in Part 3. The corresponding $AUC_{(pd,pf)}$ and $AUC_{(pf,\tau)}$ values of each part are shown in Fig. 11. It can be seen that the $AUC_{(pd,pf)}$ values increase by 0.0632 from Part 1 to Part 2, indicating that the subspace spectral angle strategy improves the detection ability. Meanwhile, from Part 2 to Part 3, the $AUC_{(pf,\tau)}$ value decreases by 0.4578, illustrating that the background suppression is improved greatly. Fig. 12 shows the box plots of the target background of the three parts in the HSI-I dataset. As shown in Fig. 12, Parts 2 and 3 lead to stronger target-background separation ability compared with Part 1. Meanwhile, the background box of Part 3 is closer to 0 and the target box span is larger than that of Parts 1 and 2.

E. Robustness Analysis

In practical application, noises inevitably exist in the original image or prior spectra, affecting the detection performance of the methods. To analyze the influence of noises on detection, different zero-mean Gaussian noises (with SNRs of 40, 35, 30, 25, 20, 15, and 10 dB) were added to the HSI-I dataset. The corresponding violin plots of $AUC_{(pd,pf)}$ for

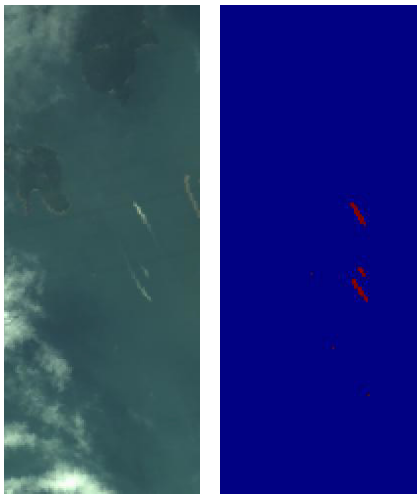


Fig. 15. Used GF-5 dataset. (a) True-color image. (b) Reference map of the target.

the 12 methods are shown in Fig. 13. It can be seen that the violin plots of CSCR, SLRMD, DDNRTD, and STSA have

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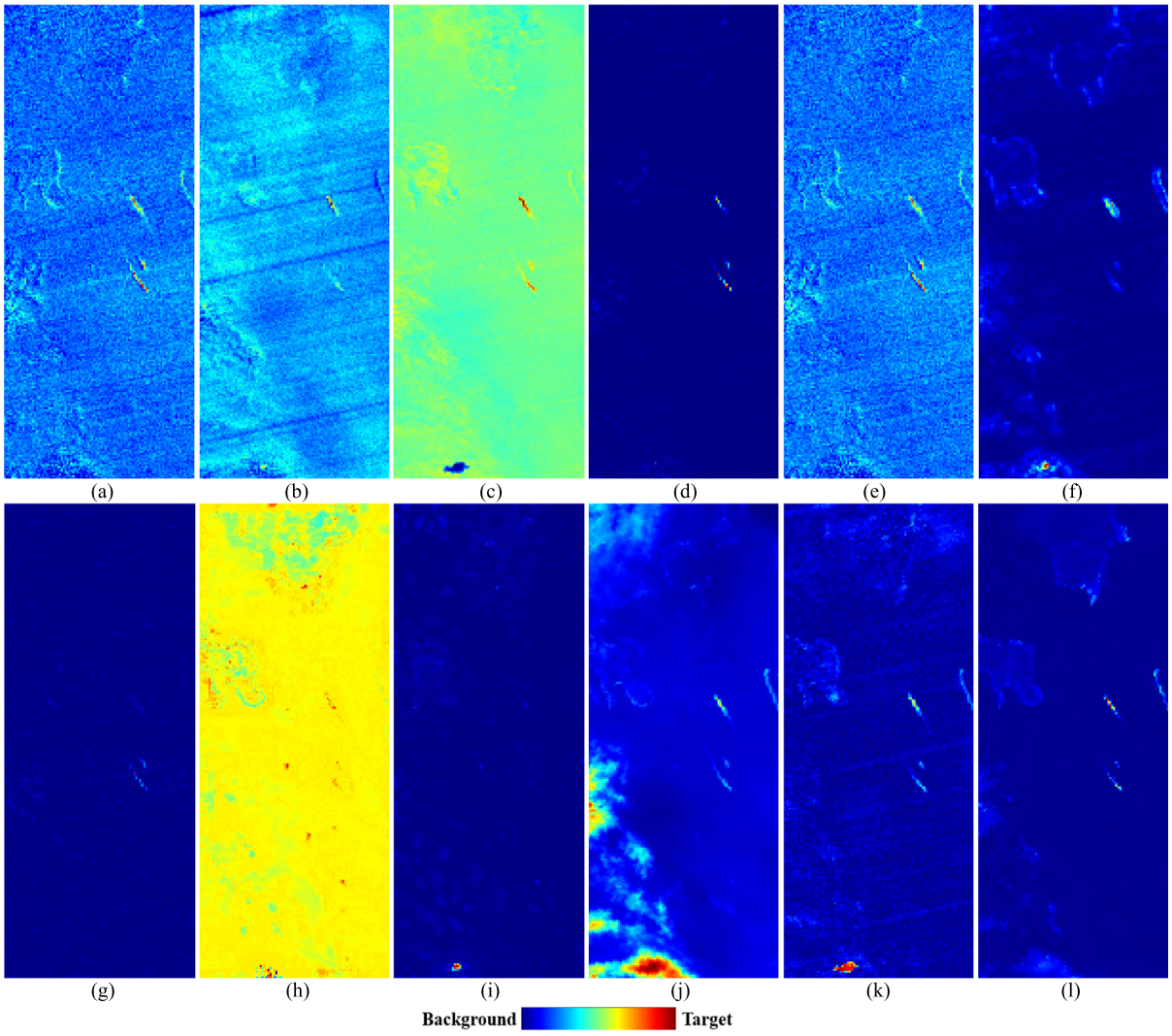


Fig. 16. Detection maps of different methods for the GF-5 dataset. (a) MF. (b) GLRT. (c) OSP. (d) OSP_GLRT. (e) CEM. (f) TPCA-CEM. (g) ACE. (h) SRBBH. (i) CSCR. (j) SLRMD. (k) DDNRTD. (l) STSA.

a smaller span and more concentrated distribution than the other eight benchmark methods, which indicates that the four methods are more robust to noises. Meanwhile, to analyze the influence of different prior target spectra on the proposed STSA method, the ten target spectra (denoted by $d1$–$d10$) in each of Figs. 2(c)–5(c) were used in turn and the results are shown in Fig. 14. It can be seen that different prior target spectra in HSI-II and HSI-III datasets have less impact on the detection performance than that for HSI-I and HSI-IV datasets, but on the whole, STSA is a method robust to different prior information.

F. Application of the STSA Method to a Larger Scene

To examine the detection performance of the methods in a larger scene, this part uses a dataset obtained from the Advanced HyperSpectral Imager (AHSI) onboard the GF-5 satellite. The study area is located in Hong Kong, China, with 150 bands and a spatial resolution of 30 m. The ships on the sea were identified as the targets to be detected in the image. Meanwhile, the backgrounds are mainly composed of complex features such as clouds, islands, and suspended solids on the sea. The true-color image and reference of the targets are provided in Fig. 15. The detection maps of the 12 methods are displayed in Fig. 16, and the corresponding AUC and SER values are listed in Table V. It can be seen that STSA is again the most accurate method in this case.

IV. Discussion

A. Uncertainty in the Prior Information

Different methods require different numbers of target or background spectra as prior information. For example, for MF, GLRT, CEM, h-CEM, ACE, and the proposed STSA method, only one target spectrum is required. OSP and OSP-GLRT require a large number of prior spectra, not only for the target, but also for the background to construct a background dictionary. For CSCR and SLRMD, they require a large number of target spectra to construct the target dictionary. Moreover, the sensitivity of each method to the prior information is

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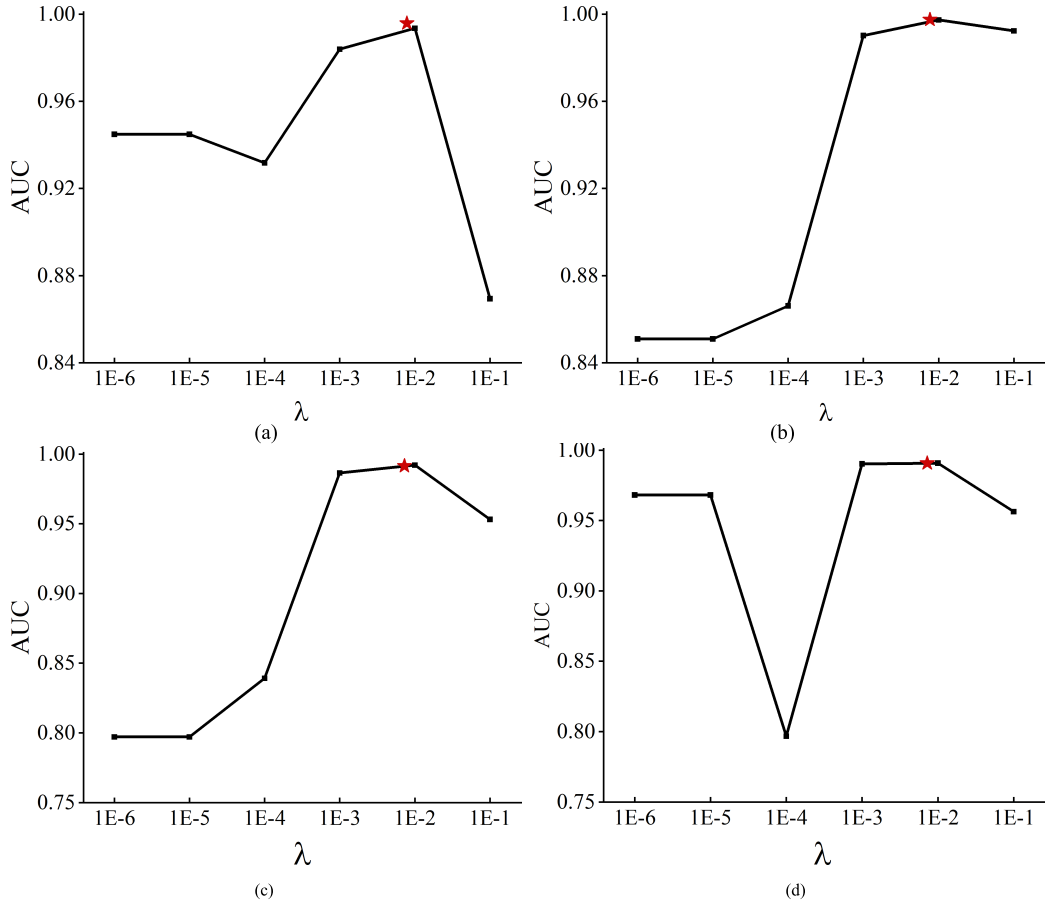


Fig. 17. $AUC_{(pd,pf)}$ values under various λ of the proposed STSA method for the four datasets. The red asterisk represents the parameter determined by $\lambda = 1/(\max(n_1, n_2)b)^{1/2}$. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

TABLE V
ACCURACY OF DIFFERENT METHODS FOR THE GF-5 DATASET

Methods	Indices			
	$AUC_{(pd,pf)}$	$AUC_{(pf,\tau)}$	AUC_{TD-BS}	SER
MF	0.8048	0.2319	0.5729	5.686
GLRT	0.6895	0.2903	0.3992	8.7166
OSP	0.9390	0.4813	0.4577	23.218
OSP-GLTR	0.8650	0.0045	0.8605	**0.2385**
CEM	0.8012	0.2366	0.5646	5.8990
TPCA-CEM	0.9209	0.0263	0.8946	0.3920
ACE	0.8452	**0.0058**	0.8394	0.2713
SRBBH	0.8394	0.6200	0.2194	38.4492
CSCR	0.7460	0.0086	0.7374	0.3267
SLRMD	0.8929	0.1106	0.7823	2.5181
DDNRTD	0.9403	0.0435	0.8968	0.5694
STSA	**0.9496**	0.0226	**0.9270**	0.3019

also different. The methods that only need a target spectrum are more susceptible to the impurity of the spectrum. However, in practical applications, it is usually difficult to obtain a large amount of prior or pure prior information. This considerably limits the practical application of the methods based on a large number of prior spectra. Currently, there are also some background reconstruction methods that can reduce the dependence on artificial priors, such as the method of constructing a local background dictionary through the dual window strategy. Therefore, for the proposed STSA method, it could be interesting to investigate whether the detection ability could be enhanced by constructing the background dictionary through the low-rank tensors.

B. Influence of the Parameters

Remote sensing images including HSIs have inherent geographical characteristics and strong spatial heterogeneity compared with natural images. Therefore, in addition to statistical methods (such as GLRT), many methods consider geospatial characteristics. For example, CSCR constructs the background dictionary based on the assumption that the target generally lacks geospatial characteristics. The potential target pixels are filtered out through the inner window, while the background pixel can be represented by a linear combination of the dual windows according to the first law of geography. SLRMD controls the sparseness of the potential target in the original image through two parameters. In general, the above methods need to introduce several parameters to enhance the geographic constraint of the method, and the selection of parameters would affect the detection ability. Here, we tested the influence of the coefficient λ of the sparse tensor $\boldsymbol{\mathcal{S}}$ in the proposed STSA method, and the results are shown in Fig. 17. It can be seen that the variation in the parameters λ can affect the detection accuracy. However, the proposed STSA method selected $\lambda = 1/(\max(n_1, n_2)b)^{1/2}$ as the empirical value according to data size (as shown by the red asterisk in Fig. 17), which is very close to the optimal parameters. Thus, the parameter

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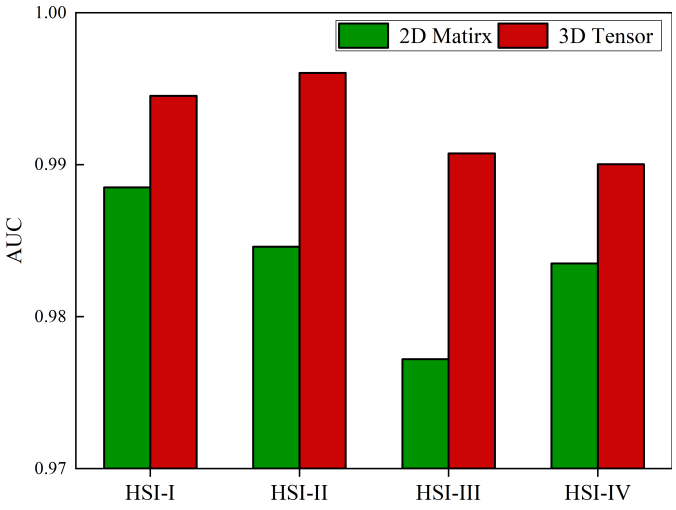


Fig. 18. Comparison between the construction strategies of the sparse model in the proposed STSA method.

can be determined automatically, where the uncertainty in parameter selection can be reduced obviously.

C. Advantage of 3-D Data-Based Decomposition

For traditional sparse models, the 3-D cube of HSI is converted into 2-D matrix data for low-rank and sparse decomposition, this step not only destroys the spatial structure of the original 3-D image but also ignores the unique geographical characteristics of remote sensing images relative to natural images, exacerbating the detection performance. For the proposed STSA method, however, the decomposition model is based on 3-D tensors. We compared the results of the above two methods of low-rank and sparse decomposition, as shown in Fig. 18. It is seen that for all datasets, the detection accuracy of the model based on the 3-D tensor is greater than that of the 2-D-based model.

D. Model Uncertainty

Each method has its unique advantages. For the mathematics-based methods, although the premise is difficult to meet under complex images, the characteristics of being simple, fast, and sensitive to targets make it have obvious advantages in the case of some large-scale images and coarse accuracy requirements. The spatial projection-based and representation-based methods have satisfactory detection ability while a large number of prior spectra can be acquired easily. Machine learning-based methods can mine deeper feature information compared with other methods.

The proposed STSA method can offset some deficiencies of the above methods to a certain extent, but it still faces several challenges. Specifically, for the decomposition model, it is difficult to completely acquire the sparse tensors from the original image. That is, a small number of sparse targets remain in the low-rank tensor, resulting in missed target detection. Therefore, how to cope with the above shortcoming will require research in the future.

V. Conclusion

In this article, we proposed the STSA method for target detection, which decomposes the original image into two parts, the low-rank tensors and the sparse tensors. The developed 3-D tensor-based decomposition model not only guarantees the geospatial structure of the original image but also reduces background interference. To cope with the inevitable background and noise contamination in the sparse tensor obtained by decomposition, the proposed STSA method applies mapping-based spectral angles and a statistical strategy for enhancement. The experimental results on four HSI datasets showed that the proposed STSA method produced greater accuracy with more satisfactory background suppression than the 11 benchmark methods.

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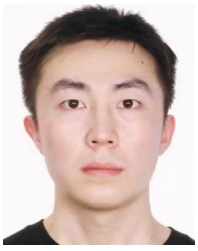
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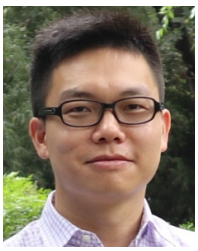
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